

How to modify your Bell Touch Tone Fone
to have 1633 cycle tone
by The Magician

In issue 62 of TAP, I gave you information on the use of 1633 cycle tones for setting up toll free loop-arounds, mass conferencing, and becoming the information operator as well as applications with Autovon.

Now I will describe how to modify a standard Bell Touch Tone fone to have the extra four 1633 buttons. An excellent article on this subject appeared in the April, 1975 issue of the now defunct "TEL" magazine. The modification is very simple and requires only a SPDT toggle switch (I recommend a miniature type) and a little hookup wire. First, unscrew the screws on the bottom of the fone and remove the cover. You will see the Touch Tone pad on a supporting bracket. Loosen the two screws on either side of the pad and gently lift it from the supporting bracket being careful not to break any wires. The pad is enclosed in a clear, thin plastic snap case. Remove the top section of the case and put it aside. Remove the bottom section and allow it to rest on the wires leading to the Touch Tone pad.

The pad (bottom up) should look like the accompanying diagram. Locate the two toroid transformers that produce the Touch Tone frequencies. They are large and donut shaped (labelled 2597 AS in diagram). Remove the protector on the left one, if present. On the bottom edge of the pad about one inch down from the toroid transformers, you will see a three pole terminal strip (labelled 1 in diagram). On some Touch Tone fones, the terminal strips are replaced by a yellow-orange wire. If you have one of these, just cut the wire in the middle, strip both ends, and these will correspond to the normally used terminals. Locate the terminal on the left and separate the two pieces of ??? metal, one on top on one on the bottom, being very careful not to break them off. Cut two one foot pieces of insulated wire and strip a little from both ends on each wire. Now take one and solder it to the bottom half of the separated terminals. Now take the other piece and solder it to the top half of the terminal, the top half being that side with the toroid transformers. Label the two wires "A" and "B" with pieces of tape with the top one being "A".

There should be two strips of five solder terminals, one on top and one on the bottom of the left toroid transformer. We are interested in the bottom one. The fourth terminal from the left (labelled 2 in diagram) is the tap for 1633 (Pa Bell uses "standardized" components for all Touch Tone encoders whether 4x4 or the normal 3x4, so the 1633 tap is there but not used). Cut another piece of wire (about one foot) and strip both ends. Solder one end of this wire to the 4th terminal, being careful that you get a good solder joint and that you are not touching any of the other terminals. Label this wire "C" and run wires "A", "B", and "C" through the bottom plastic case along with all the other wires. Replace the plastic protector on the toroid transformer (if there was one) and snap the plastic pad enclosing case back around the Touch Tone pad. Take the pad and put it back in position on its support bracket and tighten the two holding screws.

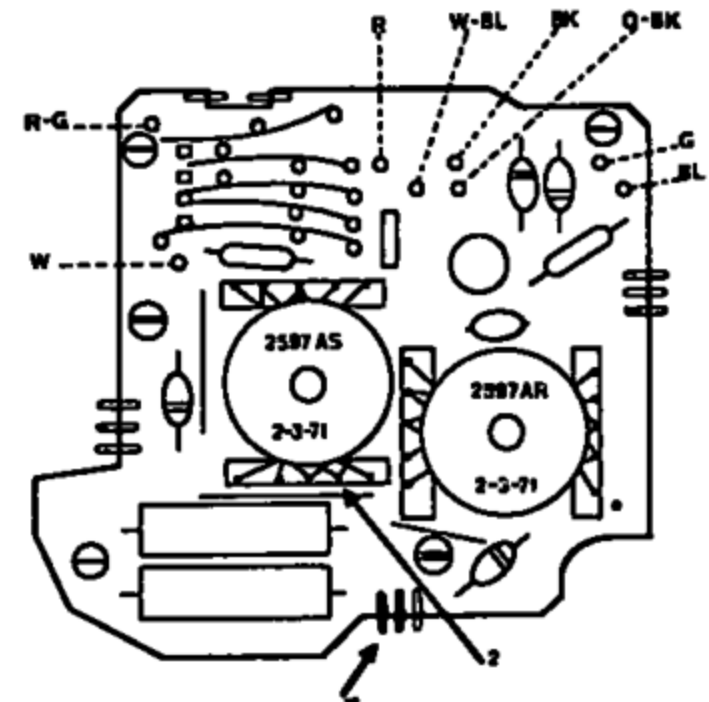
Install the SPDT switch anywhere on the telephone cover where it will not get in the way of components inside the phone. A good place has been found to be the little compartment under the hook switch on the case, looking from the back. Now take your wire labelled "A" and solder it to the middle terminal of the SPDT switch. Now take the other wires "B" and "C" and solder them to the left and right side of the switch. Put the cover back on the fone and hook up the phone to your line. If all went well, you should have the normal 3x4 touch tones with the switch in one position and when flipped to the other position, your last column of 4 buttons becomes the 1633 tones! If it doesn't work, check your solder connections to the switch and make sure you have the correct wire connected to the middle terminal of the SPDT switch. The connection to the switch is the most commonly made mistake.

When placing a regular call, remember to use the regular 1477 tone instead of 1633 or you will get a reorder. If this happens enough, Pa Bell might send a repair man to see what is wrong with your line. If you have any questions, comments, etc., please feel free to write me, The Magician, c/o TAP.



SUMMER 1980 NO. 63

1209	1336	1477	1633
697	1	2	3
770	4	5	6
852	7	8	9
941	*	0	#



PHONE'S THIRD PHON IN LA

by Cheshire

A meeting of Phone Phreaks, fancying themselves as "Communications Hobbyists" met on a quiet afternoon in January in the Los Angeles area to discuss the future of access to the telephone network by the hobbyists (phreaks). Speakers came from as far as the Midwest, and the East Coast to attend the gathering. The meeting was by invitation only, so many phreaks who live on the west coast may not even have found out that it was held.

Captain Centrex, a Midwest based phone phreak, hosted the conference. He discussed new Bell switching computers which are putting the phreaks out of the "free fones" arena of calling around the world for a dime (and getting the dime back). Cheshire Catalyst, of the New York City based TAP newsletter talked about access to the world telex network, and about equipment displayed at a telephone industry trade conference in Geneva, Switzerland he attended last September. The gist of the gathering was that computers are taking over the telephone switching world, and the phreaks had better "get digital" if they still want to rule the (micro)waves.

A few Bell System films were shown. The best of these was one called "The Bell Machine" which is highly recommended for those people interested in switching equipment. It doesn't get technical, but it is a pretty film. "Priority Immediate", a film about the fire in a New York Telephone switching center was also highly rated by the critical viewers in the audience. These films are available for civic groups, and electronic hobbyist clubs. Just call your local telephone company business office for information on how to borrow these films. Most independent telco's can get these, or other films, for you as well.

CONSUMERTRONICS CO.

TOM EDISON
TAP

LETTERS FROM READERS
2011 CRESCENT DRIVE • P.O. BOX 475
1-16-80 ALAMOGORDO, NEW MEXICO 88310

Dear Tom,

I received your last issues of TAP. Thank you very much for your placement of REBEL in the ad flyer. At least 12 new customers to date directly due to that ad, bringing our total REBEL subscriptions to just under 200 (and growing fast).

Enclosed are copies of our last three issues. You may reprint any part of them but add: "from REBEL, P.O. Box 475, Alamogordo, NM 88310." From time to time, I will insert a little blurb on TAP (free of charge) because I do want to see you survive.

I am very sorry to hear that your subscriptions are diving. I think that a big problem is that your stuff is too esoteric. I am a MSEE with plenty of applications experience and I still get lost on some of your articles. And, unfortunately, most people don't share your strong dislike for Ma Bell. The 1960s are gone - this is Looking-Out-For-No. 1 time! I have found that total obsession against the utilities has limited financial future. My main bag now is to sell SURVIVAL! I still get in my kicks against the rip-off utilities to get my rocks off but the main objective is easy and inexpensive to apply survival information. I have found that many Americans are expecting: A) A rmed Rebellion, B) Total Economic Collapse, and/or C) Foreign Power Invasion, very soon. So don't be alarmed if they don't get excited about credit card calls and red boxes.

Change your format to reflect this change in people's interests and objectives, and you will be stronger than ever, as unusual and hard-to-find info. is now very much in demand. Also, consider selling books and pamphlets of your own creations, just as I do. They do provide financial security different than that provided by a periodical.

I understand that you are considering doing your own printing. All my printing is done commercially. It is expensive. Six months ago, I invested heavily into printing equipment and supplies to do it myself - an expensive disaster. The press turned out to be a piece of junk (\$1,300!), which I've mostly repaired myself. AB DICK parts and labor are astronomical (when you can find them)! Supplies are numerous and expensive. It's a real art to work, maintain and repair offset presses, not to mention very time consuming and just plain hard work. Rubber gloves are fine until you lose a piece of paper in the form roller, which you have to dig out with your fingernails - inky hands may ruin your meals and sex life. And a press buys you very little without an offset platemaker, paper jogger, folding machine, power stapler, etc., etc. - it gets out of hand FAST!

I have found that doing my own layout work very interesting and profitable. I have a Verityper 820 Headliner and 70 fonts, a 94C Olivetti proportional typewriter (this letter) and a light table of my own construction. I can make you some headlines free of charge, just let me know and I'll send you a copy of the fonts that I have. I understand that Zenith International, 41 E. 20th. St., NYC has Veritypers at reasonable price (conditions unknown).

Well, take care. Have a good day. Keep on trucking.

Best Regards,

John J. Williams
John J. Williams, President

Visit Phun City

by Cheshire

Many people visit the Big Apple each year. Many attend the various trade shows and conferences. Others, come to catch a Broadway show or visit the other wonderful sights the City has to offer. A few come in on Friday evenings from 5 to 8 PM to share the companionship of fellow technology freaks who enjoy good food, and good beer in the back of a small restaurant on West 4th St (between 6th and 7th avenues) in world famous Greenwich Village.

Dionysia is a small Greek restaurant with a host of middle east specialties. (For those into Greek, but not Greek food, Christopher St is a block north, but don't expect us to join you) A group of TAP's writers and editors gather here in the back room, and swap information on the latest in satellite technology and telephone switching, most of which never sees print in TAP for various reasons. (lots of it is pure bull shit)

If you've wanted to come in to visit, but never had anywhere to stay, write a couple of months in advance of your visit, and we will try to arrange for you to stay with one of TAP's motly crew of regulars. Don't expect much more than a piece of floor and a sleeping bag, but you're certainly welcome to what space is available. Write to TAP at the mail drop address, and mark the envelope "Housing Committee". If your company is sending you and you just want to get together and rap, or have a local snow you around, just get in touch. A few of us have helped visiting pnoeaks spread around those expense account bucks to great advantage and to the delight of all.

Hope to see you soon.

Talk back —even to Phone Co.

LETTERS ARE RECEIVED from practically every state with similar complaints about local phone service. There's a special sense of helplessness in them, because in his or her own words each letter writer bemoans the fact that they are a "captive audience" that can't just walk across the street to a competitor phone company.

"What can we do?" they ask. But it's less of a question than a statement of unconditional surrender to whatever rate and service policies the company chooses to establish.

Like most consumers, I believe our phone system is better than any other in the world. Like an increasing number of consumers, I think it could be better than it is. Any utility, especially one which dominates an area, must be fully accountable to its customers. If a utility commission does not initiate the questioning of whether any rate schedule or company practice is in the consumer interest, then the consumers themselves must initiate that questioning and make the regulatory commission move.

That's happening in many communities where independent consumer groups which have made phone service their top (and sometimes only) priority have dramatized their complaints and won increased consumer protections. For those who despair, take heart from this success story of a consumer group in Massachusetts.

After a three-year campaign initiated by a group called Fair Share, and joined by the Public Law Institute, the Public Interest Group, the Legislative Council of Older Americans, and the attorney general's office, the Massachusetts Department of Public Utilities issued regulations earlier this year which provide increased protection against some of the most common complaints. Here's what the Massachusetts phone users won for themselves:

1. No deposits are required for most customers. Only those now in arrears, or whose service was cut off for nonpayment within the previous six months, are required to post a deposit. Customers who have already paid a deposit will have it returned—with interest.

2. Customers with complaints about any billing items can appeal to the Department of Public Utilities, which will serve as impartial arbiter.

3. Customers will have 30 days to pay a bill and if a bill is not paid on time the company must notify customer 15 days in advance if a shutoff of service is planned.

4. Customers falling behind in bill payments may arrange for installment payments over a maximum eight-month period with no interest charges.

5. Customers having difficulty paying bills because of serious illness, certified by a doctor, or for other personal emergencies for which phone service is vital, have the right to a shut off postponement of 30 days, renewable if necessary for an additional 30 days.

6. Before service can be shut off in a home where everyone is 65 or older, the company must notify the Department of Elder Affairs, and receive written permission to proceed.

7. The company must inform new customers, or old customers seeking new services, about all the possible options, beginning with the least expensive option.

8. Information on all customer rights must appear in the front of the phone book.

The gains themselves are important. But even more important is the fact that some Massachusetts consumers got together to talk back to the company and the utility commission. And they won. You might call it the Boston T (for telephone) Party. And you know what that started.

Granny's pot luck: 60 days in jail

Ventura, Calif. (UPI)—Lois Faulkner, 68, known as "Grandma Marijuana" to the schoolchildren to whom she sold and gave pot, was sentenced yesterday to 60 days in county jail.

Superior Court Judge Lawrence Storch, describing the woman's conduct as "appalling," said he limited the jail time to two months only because of her age and poor health.

Faulkner, who pleaded no contest July 31 to two counts of selling marijuana and one count of possession, told reporters outside the courtroom that she would not sell pot again but had no regrets. "Kids need marijuana to relax. It takes the kids off hard drugs," she said.

Dear Tom:

My boss, an owner of a small business is really burned with Ma Bell. He claims they double or triple the number of times he calls frequently called numbers. At 9¢ per call, they manage to get a pretty good bill worked up.

My boss once carried his complaints up into higher management levels and once was promised by someone high up that he would have an outgoing call counter installed. This never materialized and the lower levels claim no such device exists.

I would like to ask TAP readers what they know about this subject. If they don't exist, how hard would it be to develop them? There might be a pretty good market for such items.

I am an electrician and often install conduit for telephone and other communication systems. Those Bell employees are the most atrocious bunch of loafers you're ever going to see. Not only that, they've been known to sabotage the equipment of independent systems.

I have personal knowledge that Bell will ignore certain rules when they do some types of installation but make the independents follow every rule to the letter.

The Omega Man



This one has strong possibilities for the entrepreneurs: A sixteen-year-old girl obtained a bunch of carbons from credit card receipts. (These are the carbons left over after Master Charge, Bank Americard, and American Express transactions.) She had gotten the carbons from the trash bins of large department stores, and from the carbons she got essential data like card holder's name, account #, and card expiration date. Then she rented a mail address, and using the data from the carbon copies, she telephone ordered lots of salable merchandise like cameras and color TV's, billing one item to each account. The merchants sent the goods to her rented mail address without question, where it was picked up and quickly sold. This hustling little girl would probably have gotten away with the scam, but she was picked up by the police two months later on a different charge (soliciting her bottom), and when her purse was searched, they found over a hundred credit card carbons. Upon investigation the police found out from the credit card companies that many of the accounts were defrauded by fone order, and the girl was consequently charged. I got this story from the rent-a-box people, so I never got to find out why the girl was still carrying around all those scammed-out carbons.

Knowing a good opportunity when he saw it, a quick witted friend of mine was able to pick up a \$300 freebie. He had ordered an expensive microphone from Edmunds Scientific Company and had it billed to his Master Charge Card -- no signature is required for fone orders. The item was sent direct from the manufacturer, not from Edmunds as was expected, and it was sent uninsured; no signature was required to receive it. (If it was sent insured, or signature required, a non-signature -- an illegible scribble -- can be given. The illegible scribble is not a legal signature for all practical purposes, i.e. it probably wouldn't hold up in court as a signature.) After getting the package, he quickly called Master Charge to put a stop payment on the bill for the microphone, and he called the Catalogue Company, too: "Gee, I never did get that item that I ordered from your company. Are you sure that it was sent to me? I never got it. Anyway, I no longer need it, so cancel the order."

Chuck the Duck from California has discovered a new trick with airplane tickets. Recently he purchased some airplane tickets with non-sufficient funds (NSF) checks. On the tickets, in the area next to the "How Paid For," was written "Check." These tickets cannot be brought back for an immediate cash refund. So, he took a red pen that matched the ink on the tickets, and he scratched out "Check" and wrote in "Cash" next to it. He then went to a different airline and traded in the tickets that he just altered, and he received new tickets back with just "Cash" in the "How Paid For" box. The next day he brought the new tickets back for a full cash refund, which was twice as much as he was previously getting on the Black Market.

Here's another novel plane ticket scam from Chuck the Duck: Chuck called up an airline and ordered a coast-to-coast round trip ticket. He used his own name to order the ticket, but he had it sent to a friend's address. He had a fone in his name, but it was disconnected, i.e. not a working fone. The ticket, believe-it-or-not, was sent to the friend's address. It may have been just luck that the ticket was sent, but if it was due to airline practice, then I can make the following inferences: 1) the airline didn't call his fone # before sending the ticket, and if they did, they will send a ticket to a disconnected fone; 2) the airline didn't verify that he lived at the address where the ticket was being sent, and 3) the airline didn't check to see if the fone # address and the address where the ticket was being sent were the same, or it doesn't matter. Upon these assumptions, then, it may be possible to order tickets to a disconnected fone (or to any name in the fone book), and have the tickets sent to a place where they can be safely picked up, e.g. a rent-a-box or a vacant mail box.

Furthermore, airline tickets are like airline money, and they can be used with ease between different airlines; for example, bring a United ticket to American (or vice versa) and note how gladly they will accept their competitor's ticket. There are, however, a few general rules worth noting. Airline tickets are "non-transferable" and one has to say that he is the person whose name is on the ticket. Also, most airlines will not stop a ticket once it is issued, unless it is a ticket that has been reported lost or stolen. Tickets that were bought with NSF checks, and tickets that were fone ordered and not paid for, will be honored by most airlines, except for the airline that originally got beat for the ticket. So, if you are going to buy a cut-rate-ticket, ask the seller how he got it, avoid lost or stolen tickets, and use the ticket with a different airline.

FREE AIRPLANE TICKET UPDATE

In TAP Issue #57.4, JP mentioned that he didn't know if this method -- writing non-sufficient fund checks for plane tickets -- still works. Agent MDA reports that the method in #57.4 is still valid in 1979 and will probably continue to be valid as long as the airlines accept personal checks.

BLUE BOX SPEAKERS

TAP Issues 29 & 12, among others, mention using telephone earpieces for blue box speakers. Has anyone tried getting one of the little devils out of a new fortress fone lately?

First, the screw-on-cap that is over the earpiece is glued onto the handset, and it is not feasible to unscrew it by almost any amount of hand pressure.

Then, if the screw-on-cap is sawed off at the seam where it meets the handset, most likely the fine copper wire coil around the earpiece will get sawed up, too. A lot of care must be taken in sawing, or otherwise removing, the earpiece. This is not easily done out in the open at a public place. Therefore the fone handset, or the earpiece housing, should be taken home or somewhere where one can work on getting the earpiece out intact.

The telephone cannot be easily yanked away from the booth; the stainless steel cable that holds the handset to the coin box is pretty tough. The cable cannot be easily sawed, or cut with a pair of snips, but a large bolt cutter will do the job (if it is sharp).

Sawing the telephone handset in half and taking away the earpiece side is about the most practical method for those without bolt cutters. A small fine-toothed saw will cut through the handset in about a minute.

Getting the earpiece out of the handset, even at home with tools, is harder than trying to open a clam with your fingernails. After most of the plastic has been removed from the earpiece half of the handset, the remainder must be removed carefully. Melting away the plastic with a heated screwdriver, soldering iron, or other instrument is tedious; vice-grip pliers were more effective in peeling away the unwanted plastic, carefully, from around the earpiece.

The whole procedure from start to finish took an hour and a half. There must be an easier way. What do you do when you need a dozen earpieces for as many boxes? Can they be bought surplus?

M A B E L L

M is for the money that you cost me.

A is for your age, you're getting old.

B is for the business calls you carry.
There's got to be a better way I'm told.

E is for the earful not so cheerful.

L is for long distance bill's so long.

(L) This is just to tell you that I'm leaving.
Good-bye Mr Bell, hello America.

From an FCA Mericom ad.

201	093,094,101	408	293	718	204,547
202	032	412	030,042	701	133
203	020	413	009	702	176
204	456	414	088,089	703	033
205	054	415	158,159,167	704	319
206	163	416	476,478	705	477
207	003	417	157	707	223
208	039,120	418	505	712	311
209	254,289	419	057,248	713	105,151
212	017,021,023, 072,074	501	147	714	164,182
213	066,183,184	502	048	715	202,330
214	149	503	131	716	026
215	041,043	504	046	717	027,028
216	050,082	505	102	801	155
217	081,314	506	451	802	002
218	236	507	310	803	189
219	320	509	128	804	257
301	011	512	146,152	805	252,255
302	010	513	077,185	806	103
303	153	514	470,472	807	481
304	034	515	134	808	461
305	044,180	516	127	809	490 (P.R.)
306	457	517	095	812	321
307	137	518	024	813	531
308	237	519	484	814	208,307
309	203,260,277	601	059	815	087
312	086,097,098, 196,234	602	064	816	144
313	083,096	603	004	817	150
314	143,177,251	604	493	819	474
315	303	605	138	901	187
316	140	606	317	902	452
317	080	607	025	904	056
318	051	608	201,329	906	295
319	312	609	101	907	492
401	019	612	126	912	316
402	139	613	473	913	145
403	454	614	079	914	069
404	035,063	615	047	915	221
405	148,215	616	084	916	160
406	154	617	001,006,007, 008	918	141
				919	036

And not too often.



PETRODOLLARS, NICKELS AND QUARTERS are choking Findlay Ohio, pay phones. When the 30 Saudi Arabian students enrolled at Findlay College get homesick, they call home, at the rate of \$9.65 for the first three minutes and \$3.20 for each additional minute. That means \$44.05 worth of change for the average 20-minute Saudi chat. The coinage keeps clogging the collection boxes on Findlay's 23 pay phones, knocking them out of order.